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Quartz-Fluid Interfacial Tension of Hydrogenated and De-Hydrogenated Organic Hydrogen Carriers: Implications for Hydrogen Geological Storage

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Abstract

A hydrogen-based economy necessitates effective hydrogen storage, and Organic Hydrogen Carriers (OHCs) represent a promising solution to this challenge. OHCs offer a practical, safe, and efficient means for handling, distributing, and storing hydrogen, particularly in geological formations. Their performance in subsurface environments, however, depends significantly on interfacial properties at the pore scale—especially the equilibrium contact angle (θ_E) and interfacial tensions between solid-brine (γ_{SL}) and solid-OHC (γ_{SOHC}) interfaces. These parameters influence how OHCs interact with the rock matrix and determine their storage capacity and confinement stability. Due to the technical difficulty of direct measurement, these interfacial properties are often calculated using theoretical models such as Young's and Neumann's equations.

In this study, interfacial parameters were theoretically determined for a quartz substrate in contact with brine and two OHC phases: toluene (de-hydrogenated form) and methylcyclohexane, MCH (hydrogenated form). The analysis was conducted across reservoir-relevant conditions (temperatures of 298–343 K, pressures of 1–20 MPa, and 1 M NaCl salinity), using observed advancing and receding contact angles and fluid-fluid interfacial tension (IFT) as input.

Results revealed that with increasing pressure, the contact angle θ_E increases while γ_{SOHC} decreases, indicating enhanced wettability and reduced IFT due to stronger interactions between OHC molecules and quartz surfaces. Conversely, γ_{SL} remains largely unaffected by pressure changes due to the minor density variation of brine. As temperature rises, both θ_E and γ_{SL} increase, while γ_{SOHC} decreases. These temperature and pressure trends influence how efficiently OHCs can be confined and distributed within reservoir pores. Additionally, toluene demonstrated higher θ_E and lower γ_{SL} and γ_{SOHC} values than MCH under similar conditions, attributable to its higher density. For instance, at 343 K and 5 MPa, toluene has a density of 0.82478 g/ml compared to 0.73623 g/ml for MCH.

These findings highlight the potential of quartz-rich sandstone formations for secure and efficient OHC-based hydrogen storage. Optimizing interfacial interactions between OHCs and geological substrates is crucial for advancing hydrogen geo-storage technologies and supporting the transition to sustainable energy systems.

Keywords: Organic Hydrogen Carriers, Hydrogen Geo-storage, Thermo-physical Parameters, Salinity, Fluid-Rock Interfacial Tension

Introduction

Rising energy consumption is a direct result of the exponential expansion of the world's population and industrial output in recent decades [1–4]. The primary energy source for a long time, fossil fuels, are now being used up faster than they were discovered and are also a major contributor to climate change [5–7]. The use of underground hydrogen storage (UHS) and carbon dioxide geo-storage (CGS) both provide practical ways to reduce carbon emissions and meet energy demands [8–11]. In contrast to UHS, which enables the effective conversion of renewable energy excess into hydrogen (H_2) by electrolysis, which permits energy storage and usage, CGS allows the safe storage of CO_2 inside geological formations [12–16]. These technologies work together to make our energy future more secure and less polluting [17, 18].

The increasing relevance of hydrogen in fuel cells, cars, and refineries makes UHS a viable option for creating a sustainable energy system centered on hydrogen. However, there is still a big problem in storing hydrogen on a massive scale [19, 20]. A new, safe, and adaptable way to store hydrogen has emerged with the advent of organic hydrogen carriers (OHCs) [21–23]. OHCs are ideal for real-world use because they are thermally stable, have outstanding hydrogenation characteristics, and operate like fuel when exposed to air [22, 24]. Nevertheless, the rock-fluid and fluid-fluid interfaces interfacial tension (IFT) and rock-OHC-brine wetting behavior, i.e., important thermo-physical and petro-physical factors, significantly impact the efficiency of UHS [24–26]. When preserving fluids during storage after injection, their distribution and withdrawal of these elements are crucial [26–28].

Problems occur when measuring contact angles for systems such as rock-H_2-brine. These problems are caused by the tendency of metals to embrittle, the reactivity of hydrogen gas, and the excellent compressibility of hydrogen gas [29]. Rock-OHC-brine systems, on the other hand, are safer and easier to operate because measuring contact angles is less complicated [22, 26]. The experimental settings provide valuable information on the behavior of multiphase systems and their interactions with surfaces. Laboratory settings have extensively used advancing (θ_a) and receding (θ_r) contact angles, as well as IFT of fluid-fluid interactions ($\gamma_{Liq\text{-}Liq}$), to evaluate wettability for rock-fluid and fluid interactions in porous systems [30, 31]. This is mainly because these parameters are very straightforward to measure [22, 25]. Currently, there are no experimentally attainable methods for measuring IFTs involving solid surfaces, such as solid-liquid (γ_{SL}) and solid-OHC (γ_{SOHC}) [24, 32, 33].

Rock–fluid IFTs are frequently estimated indirectly through fluid-fluid IFTs, contact angle data, and theoretical models, including Young's equation or surface energy-based methods. Also, some scholars have used a semi-empirical method that merges Young's and Neumann's equations of state [24, 33, 34]. Much research on brine and gases (CO_2 and H_2) is available to measure IFT and rock-wetting behavior under geological storage situations. However, when it comes to storing hydrogen through OHCs, a detailed investigation into the associated properties, such as the petrophysical and the thermophysical, is still lacking and requires attention.

Filling this knowledge disparity and to shed light on the fundamental interactions for geological hydrogen storage, we methodically computed the IFT values for the quartz-OHC-brine case. To evaluate the wetting behavior and OHC storage efficiency in porous media, the consequence of varying pressures and temperatures on the equilibrium contact angles (θ_E) is explicitly investigated in this research. A complete data set concerning rock-fluid IFT, encompassing the solid-liquid (γ_{SL}) and solid-OHC (γ_{SOHC}) IFTs, is offered. In subsurface fluid storage, these interactions control the fluid distribution dynamics, the retaining potential, and the withdrawal; thus, they are of utmost importance. Our discoveries improve knowledge of hydrogen geo-storage processes and provide the groundwork for OHCs' broad application in sustainable energy storage.

Experimental

Materials

Quartz specimens were obtained from Ward's Science, USA, to measure contact angles (wettability) in this research. Thermo Fisher Scientific provided 99% pure methylcyclohexane (MCH – OHC hydrogenated form), while high-performance liquid chromatography (HPLC) grade toluene (the dehydrogenated form) was obtained from VWR chemicals. Sigma Aldrich provided 0.999 mol % pure Sodium chloride (NaCl).

Using deionized (DI) water having 0.02 mS/cm electrical conductivity, a one-molar solution of NaCl was prepared to create the liquid phase. Interfacial and wetting behavior for the quartz-OHC-brine system could be accurately evaluated owing to the regulated solution composition that maintained consistent experimental conditions.

Methodology

To measure the fluid-fluid IFT and the contact angle, Core Lab's built Hastelloy cell, capable of withstanding temperatures ~ 523 K and pressures > 69 MPa, was used [22, 35]. In this investigation, the data (IFT and the contact angle) was collected using geo-storage conditions with temperatures 303 to 343 K and 1 to 20 MPa pressures. The pendant drop approach was used to evaluate the IFT of brine in the OHC environment, also known as $\gamma_{OHC\text{-}Brine}$. The experimental circumstances, the kind of OHC, brine density, and pendant drop diameter administer the IFT [36]. Ali et al. [22, 26] provided a complete overview of the procedures for determining brine-OHC IFT.

To replicate the reservoir environments, thin samples of polished quartz were aged for 7 days at 323 K in relevant OHCs, and ultra-pure nitrogen gas was utilized to remove surface contaminants. To estimate the leading (θ_a) and trailing (θ_r) dynamic contact angles of the rock-OHC-brine arrangement, the 17° tilted plate approach was used right before the droplets migrated at the advancing and receding edges, respectively [37]. Images were captured and processed using ImageJ; the mean θ_a and θ_r values were determined from three independent experiments at similar reservoir conditions. To measure the contact angles and IFT, we detailed our previous works' exact setup, necessary equipment, and procedure [22, 38, 39].

We calculated the θ_E using Eq. 1 and Eq. 2, using IFT at the OHC-brine interface and θ_a and θ_r values for the quartz-OHC-brine system. The equilibrium contact angle (θ_E) is the same as the one that represents Young's equilibrium, described by Young (θ_Y) [40].

$$\theta_E = arcos\left(\frac{N_a cos\theta_a + N_a cos\theta_r}{N_a + N_r}\right) \tag{1}$$

$$N_a = \left(\frac{sin^3\theta_a}{2 - 3cos\theta_a + cos^3\theta_a}\right)^{\frac{1}{3}} \text{and } N_r = \left(\frac{sin^3\theta_r}{2 - 3cos\theta_r + cos^3\theta_r}\right)^{\frac{1}{3}} \tag{2}$$

Young's equation (Eq. 3) presents the balance of the intermolecular forces, i.e., $\gamma_{OHC\text{-}Brine}$, γ_{SL}, and γ_{SOHC}. Furthermore, IFT at solid-fluid interface estimation is possible using Neumann's equation of state (Eq. 4) [41].

$$cos\ \theta_E = \frac{\gamma_{\text{SOHC}} - \gamma_{\text{SL}}}{\gamma_{\text{OHC−Brine}}} \tag{3}$$

$$\gamma_{\text{SOHC}} = \gamma_{\text{SL}} + \gamma_{\text{OHC-Brine}} - 2\sqrt{\gamma_{\text{SLx}}\gamma_{OHC-Brine}}\left(1 - \beta\left(\gamma_{\text{SL}} - \gamma_{\text{OHC-Brine}}\right)^2\right) \tag{4}$$

The simplified versions of equations that result from the combination of Eq. (3) and (4) are denoted by the Eq. (5) and (6).

$$cos\,\theta_E = 1 - 2\sqrt{\frac{\gamma_{\text{SL}}}{\gamma_{\text{OHC-Brine}}}}\left(1 - \beta\left(\gamma_{\text{SL}} - \gamma_{\text{OHC-Brine}}\right)^2\right) \tag{5}$$

$$1 - \left(\frac{1 - cos\,\theta_E}{2}\right)\frac{\sqrt{\gamma_{\text{OHC-Brine}}}}{\sqrt{\gamma_{\text{SL}}}} = \beta\left(\gamma_{\text{SL}} - \gamma_{\text{OHC-Brine}}\right)^2 \tag{6}$$

Therefore, to get β and γ_{SL} Eq. (6) will be solved using the regression approach- the two-variable least-square technique. The Eq. (6) is in parabola form. Once the β and γ_{SL} have been determined, the γ_{SOHC} can be calculated using Eq. (4).

Results and Discussion

Temperature and pressure effects on equilibrium contact angle

Wettability is essential when assessing a geological formation's capacity to store fluids [31, 38]. A lower storage capacity and a greater danger of fluid leakage are the results of capillary forces decrease owing to contact angle increase, as stated in the Young-Laplace equation [42, 43]. Intermolecular forces and fluid density influence the system's wettability (θ_E), which fluctuates as a function of pressure [22, 26, 44]. In general, pressure affects wettability because it increases fluid density and amplifies the contact between the fluid and the rock surface [22, 26]. The existence of intermolecular forces changes the interfacial energy, which in turn affects θ_E, and this effect is temperature dependent [24, 31]. The effect of pressure and temperature on θ_E in quartz-OHC-brine systems is shown in Figure 1. A similar pattern is displayed by both toluene and MCH cases, with values increasing with pressures and temperatures. An understanding of these relationships is essential to maximize the efficiency of hydrogen geo-storage utilizing OHCs.

Comparing toluene-quartz with MCH-quartz in the same experimental setting, the former shows a broader range with greater θ_E values. To say, at 15 MPa and 323 K, toluene records higher (θ_E = 106.65°) than for MCH (θ_E = 97.75°). Both OHCs θ_E increased at constant pressure, although toluene-quartz was more sensitive to temperature changes than MCH-quartz. For instance, at 20 MPa θ_E increased by 8.25° (from 103.65° at 298 K to 111.9° at 343 K) for toluene-quartz than MCH-quartz by 4.5 ° (from 97.75° at 298 K to 102.25° at 343 K). The interactions between solid and liquid may be inferred from θ_E change with temperature rise. Similarly, in both instances, θ_E increases with increasing pressure, suggesting that the wettability resistance is strengthened; however, the increase in θ_E for toluene-quartz is more rapid. For example, at 323 K θ_E increased by 11.25° (from 98.1° at 1 MPa to 109.35° at 20 MPa) for toluene-quartz than MCH-quartz by 9.5° (from 90.2° at 1 MPa to 99.7° at 20 MPa).

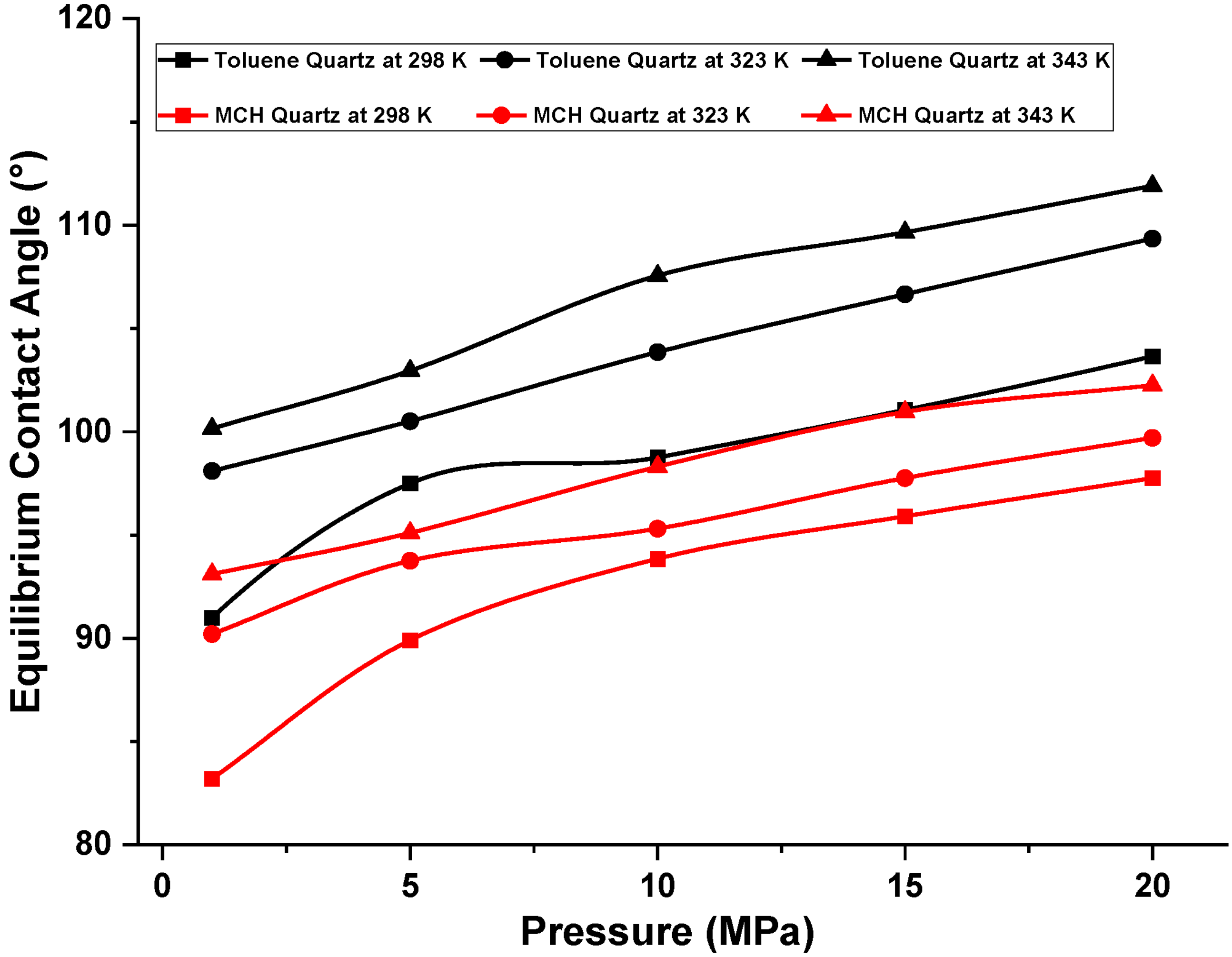

Figure 1—Equilibrium contact angle (θ_E) influenced by varying pressure and temperature for the quartz-OHC-brine system.

Unlike usual trends [25], this study found that θ_E increased as pressure and temperature rose because of changes in the interfacial free energy. Specifically, toluene, which has a minor density difference ($\Delta\rho$) concerning brine than MCH, has larger θ_E and effectively wets the surface of quartz more than MCH [45]. This suggests that higher temperatures decrease quartz's affinity for brine by weakening interactions between quartz and water and enhancing its OHC-wet characteristics.

Temperature and pressure effects on rock-brine IFT

Minimal variations are seen in the intermolecular exchanges involving fluid (brine) and quartz at different pressures, as seen in Figure 2, since brine density is relatively constant. The quartz-brine IFT (γ_{SL}) continues to be unvarying concerning pressure rise. However, γ_{SL} increases with temperature increase. For instance, in the toluene case at 15 MPa, γ_{SL} increases to 15.51 mN/m at 343 K from 13.32 mN/m at 298 K and at the same pressure condition γ_{SL} in MCH environment increases to 16.72 mN/m at 343 K from 15.34 mN/m at 298 K. With temperature increase, for toluene environment the change in γ_{SL} is noticeable compared to MCH surrounding. A change of ~ 2.12 mN/m and ~1.38 mN/m is observed for toluene and MCH at the aforementioned reservoir conditions.

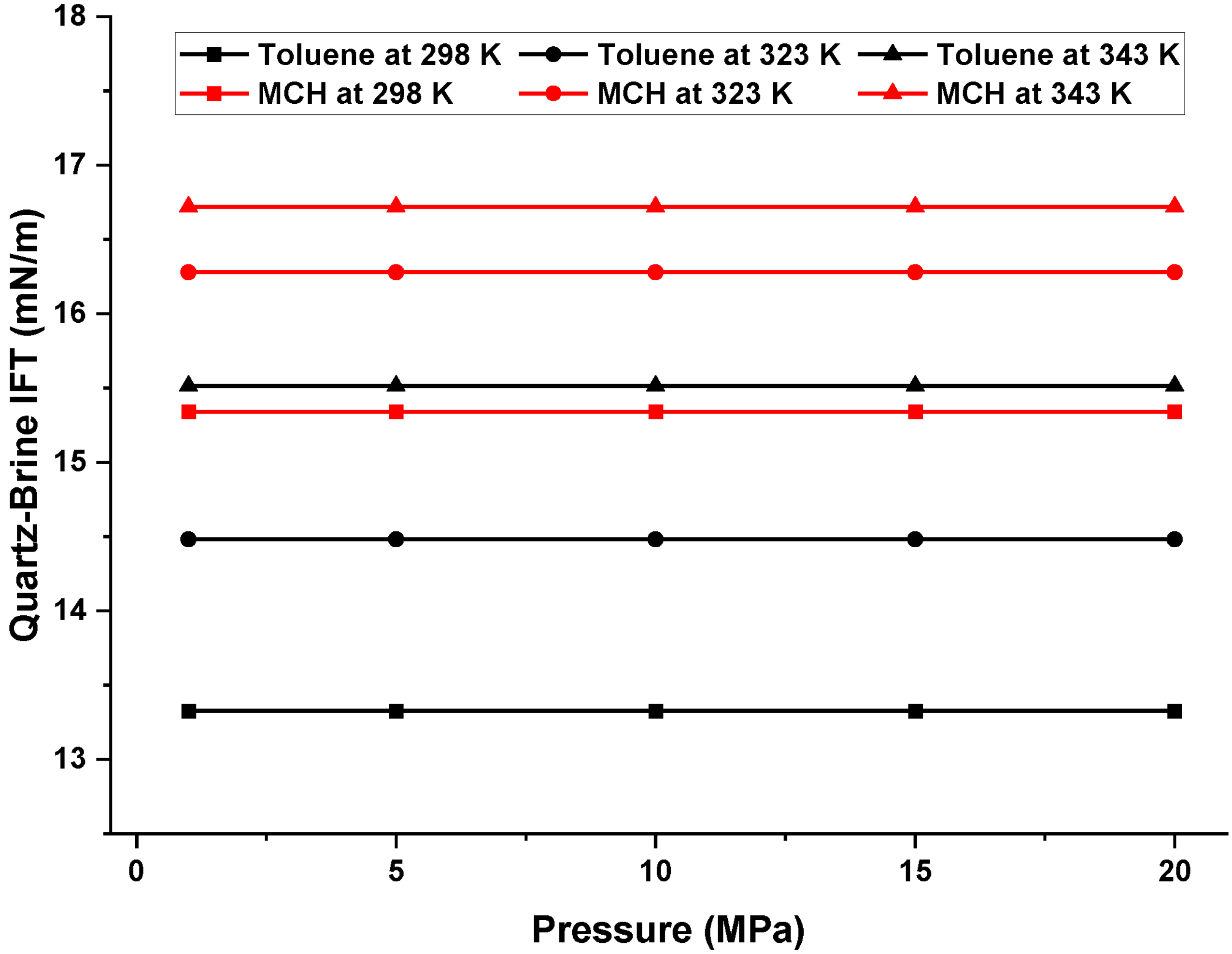

Figure 2—Pressure and temperature impact the interfacial tension (γ_{SL}) between brine and quartz.

This peculiar behavior of increase in γ_{SL} with temperature rise is because of the surrounding fluid characteristics (specifically density) [22, 26] and the enhanced OHCs adsorption onto quartz substrate (as demonstrated by θ_E values), which hinders brine interaction with the substrate surface. Furthermore, the intermolecular force in volatile organic liquids is temperature-dependent, leading to heightened molecular mobility with effective interactions between the organic phase and rock.

Effect of pressure and temperature on rock-OHC IFT

One excellent option for large-scale hydrogen storage and delivery is organic hydrogen carriers (OHC) like MCH and toluene [24]. These carriers provide a solution for reversible hydrogen storage via sequences of hydrogenation and dehydrogenation [22, 26]. Volume loss under pressure is more pronounced for OHCs than water because of lower van der Waals forces, in contrast to water's strong hydrogen bonding and relative incompressibility. Having far greater compressibility of toluene and MCH than water, their densities rise with increasing pressure due to molecular compression [22]. OHCs pressure-dependent density and compressibility variations knowledge are vital for efficient hydrogen storage and leak prevention. Owing to OHC's greater compressibility, they can adapt to changes in pressure more effectively, which impacts their characteristics such as wetting behavior, interfacial characteristics and phase properties under storage circumstances of subsurface. These characteristics are essential to store hydrogen in geological sites as they govern fluid retention, distribution and effectiveness of fluid extraction in formations [37, 46–48].

The quartz-OHC interfacial tension (γ_{SOHC}) variation as a function of pressure and temperature is shown in Figure 3. γ_{SOHC} decreases for both MCH and toluene with an increase in pressure and temperature.

For instance, at 323 K, with pressure increase to 20 MPa from 1 MPa leads to a decrease in γ_{SOHC} by ~ 2.36 mN/m for toluene (from 6.97 mN/m to 4.60 mN/m) and for MCH by ~ 2.84 mN/m (from 13.28 mN/m to 10.43 mN/m). Increasing cohesive energy density, which decreases γ_{SOHC} as pressure increases, improves the adherence of MCH and toluene to quartz and eventually promotes quartz-MCH or quartz-toluene hydrophobic wettability [33]. Contrarywise, at a constant pressure of 15 MPa, with a temperature rise to 343 K from 298 K, causes γ_{SOHC} drop by ~ 2.72 mN/m for toluene (from 7.18 mN/m to 4.46 mN/m) and for MCH by ~ 3.09 mN/m (from 12.19 mN/m to 9.10 mN/m). As previously noted, the temperature-driven shift in intermolecular forces boosts molecular mobility and strengthens OHCs and quartz rock interactions.

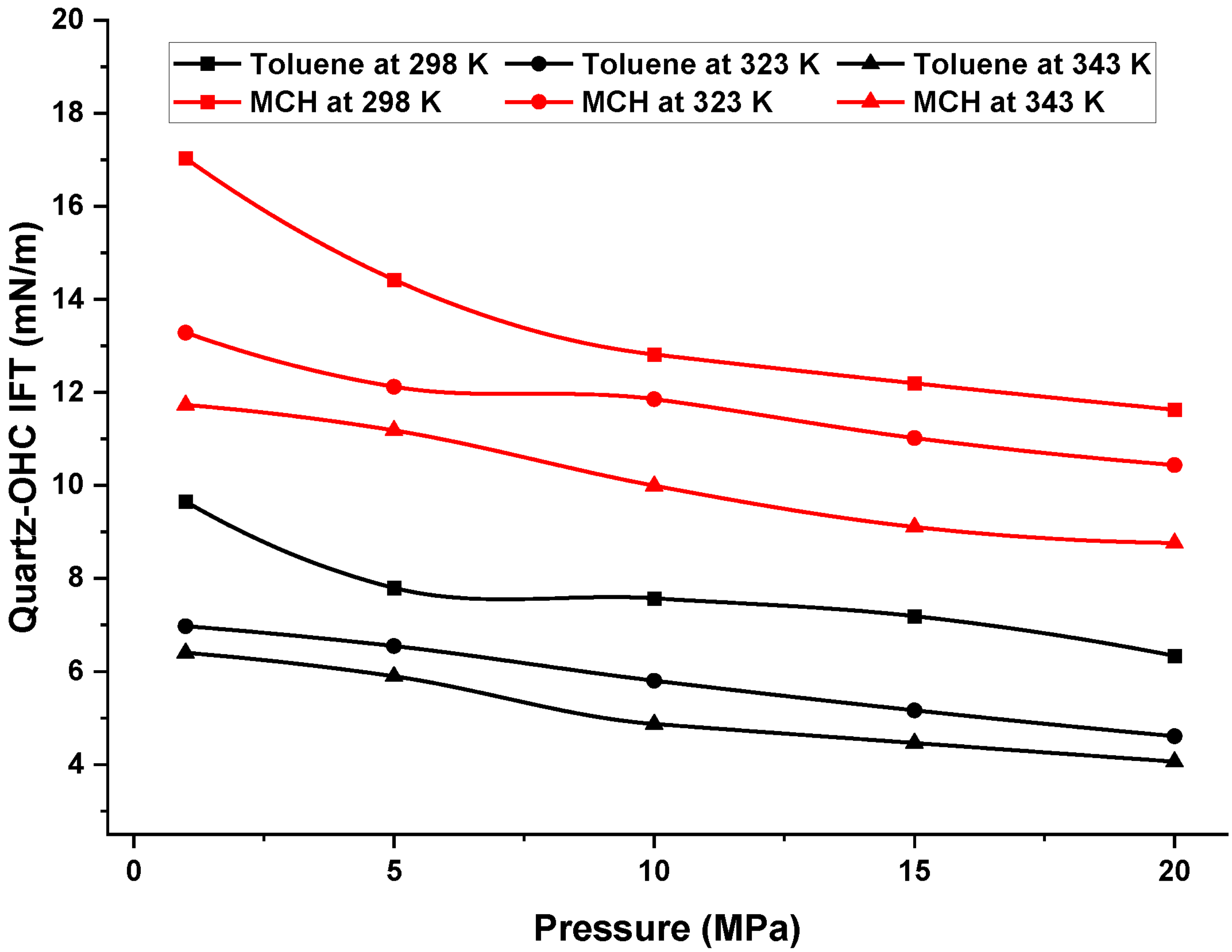

Figure 3—Quartz-OHC interfacial tension (γ_{SOHC}) variation as a function of pressure and temperature.

As illustrated in Figure 3, under identical thermophysical circumstances, MCH always displays a greater γ_{SOHC} than toluene because of its lower density than toluene and less contact with the surface of the quartz [22, 33]. A rise in pressure for a specific fluid and a decrease in γ_{SOHC} is caused by an increase in fluid density and an enhancement of intermolecular forces between the fluid and the rock [49]. For example, at 15 MPa and 298 K, for toluene and MCH, the respective calculated γ_{SOHC} values are 7.18 mN/m and 12.19 mN/m, respectively. Given that toluene has a larger density than MCH, the γSOHC value for MCH always stays higher than that of toluene, indicating an inverse trend.

Conclusion

Organic hydrogen carriers provide a potential option for practical hydrogen storage, which is essential for the hydrogen economy. Hydrogen is a possible energy source for large-scale applications due to OHC technology, allowing for easy handling, transport, and safe geological storage. The interfacial properties, including wetting properties (equilibrium contact angle; θ_E) and interfacial tensions (rock-brine; γ_{SL}, and rock-OHC; γ_{SOHC}) play a key role in regulating injection and distribution of fluid, and storage and retrieval while demand is more. To fill the gap in the literature caused by experimental restrictions, this study employs a semi-empirical technique to predict these critical factors for the quartz-OHC-brine system at settings of temperature range = 298 – 343 K and pressure range = 1 – 20 MPa (reservoir physio-thermal environments).

This study's findings show that the reservoir physio-thermal environments significantly affect wettability as the equilibrium contact angles (θ_E) rise with increasing pressure and temperature. The quartz–brine interfacial tension (γ_{SL}) was unaffected by pressure since brine density is relatively constant and does not vary much with pressure. However, the temperature increase led to a rise in γ_{SL} for quartz-toluene-brine and quartz-MCH-brine systems. On the contrary, increasing pressure and temperature resulted in a decrease in γ_{SOHC}. The density difference between the two fluids was the main reason, under the same circumstances, the hydrogenated OHC form (MCH) showed greater rock-fluid interfacial tensions (γ_{SOHC} and γ_{SL}) than the dehydrogenated OHC form (toluene). These findings highlight the importance of interfacial characteristics in subsurface hydrogen storage utilizing OHCs.

To conclude, it is crucial to comprehend and optimize the interfacial dynamics involving brine, OHCs, and rocks to maximize the efficiency of subsurface hydrogen storage. This research lays the groundwork for future hydrogen geo-storage using organic hydrogen carriers (OHCs), which will aid in the shift to a carbon-free energy future. Improving the safety and scalability of hydrogen storage methods will require ongoing study of the petro- and thermo-physical characteristics of OHCs under different geological environments.

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